

Math 105: Calculus with Algebra
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quiz archive

Gradescope code: **B5J8EN**

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- (1) Solve the equation $|x - 2| = 7$ for x .

Solution. Since $|x - 2| = 7$, either $x - 2 = 7$ or $x - 2 = -7$. Thus $x = 9$ or $x = -5$.

- (2) Solve the inequality $|x + 3| < 4$.

Solution. The inequality $|x + 3| < 4$ is equivalent to $-4 < x + 3 < 4$. Subtracting 3 throughout gives $-7 < x < 1$.

Thus the solution is $(-7, 1)$. On the number line, shade the interval between -7 and 1 , with open circles at -7 and 1 .

- (3) Express

$$1 + \frac{1}{\frac{x}{x+2}}$$

as a single fraction.

Solution. We have

$$1 + \frac{1}{\frac{x}{x+2}} = 1 + \frac{x+2}{x} = \frac{x+x+2}{x} = \frac{2x+2}{x}.$$

Thus the single fraction is $\frac{2x+2}{x}$.

The original expression is defined only when $x \neq -2$ and $x \neq 0$.

(1) Consider the line

$$x + 2y = 2.$$

(a) To find the slope, solve the equation for y :

$$x + 2y = 2$$

$$2y = -x + 2$$

$$y = -\frac{1}{2}x + 1.$$

Thus the slope is

$$\boxed{-\frac{1}{2}}.$$

(b) A line parallel to this line has the same slope, so its slope is $-\frac{1}{2}$. Since it passes through $(2, -1)$, we can use point-slope form:

$$y - (-1) = -\frac{1}{2}(x - 2).$$

Thus

$$y + 1 = -\frac{1}{2}x + 1,$$

and hence

$$\boxed{y = -\frac{1}{2}x}.$$

(c) A perpendicular line has slope equal to the negative reciprocal of the original slope. Since the original slope is $-\frac{1}{2}$, the perpendicular slope is

$$\boxed{2}.$$

(2) Let

$$f(x) = 3x^2 + 2.$$

We want to compute

$$\frac{f(a+h) - f(a)}{h}.$$

First,

$$f(a+h) = 3(a+h)^2 + 2.$$

Expanding gives

$$f(a+h) = 3(a^2 + 2ah + h^2) + 2 = 3a^2 + 6ah + 3h^2 + 2.$$

Also,

$$f(a) = 3a^2 + 2.$$

Therefore

$$\begin{aligned} \frac{f(a+h) - f(a)}{h} &= \frac{3a^2 + 6ah + 3h^2 + 2 - (3a^2 + 2)}{h} \\ &= \frac{6ah + 3h^2}{h} \\ &= \boxed{6a + 3h}. \end{aligned}$$

(3) Let

$$g(x) = \begin{cases} x^2, & \text{if } x \leq 4, \\ -|x - 4|, & \text{if } x > 4. \end{cases}$$

(a) To compute $g(0)$, notice that $0 \leq 4$, so we use the first part of the definition:

$$g(0) = 0^2 = \boxed{0}.$$

Since $5 > 4$, we use the second part to compute $g(5)$:

$$g(5) = -|5 - 4| = -|1| = \boxed{-1}.$$

(b) For $x \leq 4$, the graph is the parabola

$$y = x^2.$$

Since $x = 4$ is included, there is a closed point at

$$(4, 16).$$

For $x > 4$, we have $x - 4 > 0$, so

$$|x - 4| = x - 4.$$

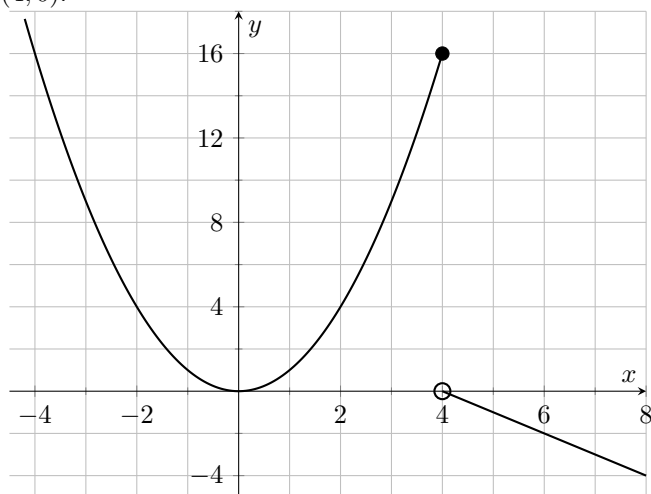
Therefore

$$-|x - 4| = -(x - 4) = 4 - x.$$

Thus the second piece is the line

$$y = 4 - x, \quad x > 4,$$

with an open point at $(4, 0)$.



The first piece covers all $x \leq 4$ and the second covers all $x > 4$. Therefore the domain is

$$\boxed{(-\infty, \infty)}.$$

The parabola gives every $y \geq 0$, while the line gives every $y < 0$. Together these give every real number.

Therefore the range is

$$\boxed{(-\infty, \infty)}.$$

- (1) The graph of $y = f(x)$ is shown below, and we want to sketch

$$y = f(x - 1) + 2.$$

The expression $f(x - 1)$ shifts the graph of f **one unit to the right**. Then adding 2 shifts the graph **two units up**.

Thus every point

$$(x, y)$$

on the original graph moves to

$$(x + 1, y + 2).$$

In particular,

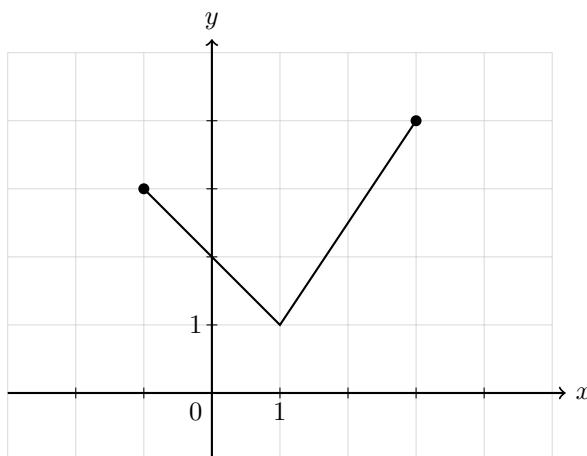
$$(-2, 1) \mapsto (-1, 3),$$

$$(0, -1) \mapsto (1, 1),$$

and

$$(2, 2) \mapsto (3, 4).$$

Therefore the transformed graph is:



Shift right 1 unit and up 2 units.

- (2) Let

$$f(x) = \sqrt{x+1} \quad \text{and} \quad g(x) = x^2 - 2.$$

We want to compute

$$(f \circ g)(2).$$

Recall that

$$(f \circ g)(2) = f(g(2)).$$

First compute the inside function:

$$g(2) = 2^2 - 2 = 4 - 2 = 2.$$

Then

$$f(g(2)) = f(2) = \sqrt{2+1} = \sqrt{3}.$$

Therefore

$$(f \circ g)(2) = \sqrt{3}.$$

(3) Find

$$\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}.$$

If we substitute $x = 3$ immediately, we get

$$\frac{3^2 - 9}{3 - 3} = \frac{0}{0},$$

so we need to simplify first.

Factor the numerator:

$$x^2 - 9 = (x - 3)(x + 3).$$

Therefore, for $x \neq 3$,

$$\frac{x^2 - 9}{x - 3} = \frac{(x - 3)(x + 3)}{x - 3} = x + 3.$$

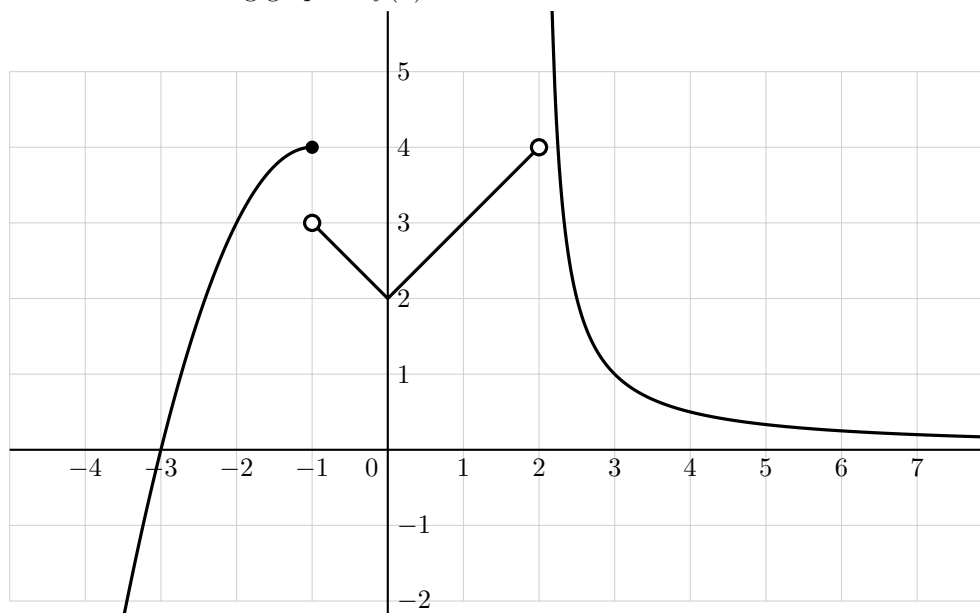
Now we can take the limit:

$$\lim_{x \rightarrow 3} (x + 3) = 3 + 3 = 6.$$

Thus

$$\boxed{\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} = 6}.$$

- (1) (5 points) Consider the following graph of $f(x)$.



- (a) As x approaches 2 from the right, the graph rises without bound. Therefore

$$\lim_{x \rightarrow 2^+} f(x) = +\infty.$$

- (b) As x approaches 2 from the left, the graph approaches the open point $(2, 4)$. Therefore

$$\lim_{x \rightarrow 2^-} f(x) = 4.$$

- (c) The right-hand limit is $+\infty$, while the left-hand limit is 4. Since the one-sided limits do not agree,

$$\lim_{x \rightarrow 2} f(x) \text{ does not exist.}$$

- (d) As $x \rightarrow -1^-$, the graph approaches 4, while as $x \rightarrow -1^+$, the graph approaches 3. Since the one-sided limits are different,

$$\lim_{x \rightarrow -1} f(x) \text{ does not exist.}$$

- (e) Near $x = 0$, the graph lies on the two line segments meeting at $(0, 2)$, so from both sides $f(x)$ approaches 2. Thus

$$\lim_{x \rightarrow 0} f(x) = 2.$$

(2) (1 point)

$$\lim_{x \rightarrow -3} \frac{x+3}{x^2+x+1}.$$

The denominator is nonzero at $x = -3$, since $(-3)^2 + (-3) + 1 = 7$. Therefore direct substitution gives

$$\boxed{0}.$$

(3) (1 point)

$$\lim_{x \rightarrow 3} \frac{x-7}{3-x}.$$

As $x \rightarrow 3$, the numerator approaches -4 . If $x \rightarrow 3^-$, then $3-x \rightarrow 0^+$, so the quotient approaches $-\infty$. If $x \rightarrow 3^+$, then $3-x \rightarrow 0^-$, so the quotient approaches $+\infty$. Therefore

the limit does not exist.

(4) (1 point)

$$\lim_{x \rightarrow -3} \frac{x+3}{x^2-2x-15}.$$

Factor the denominator:

$$x^2 - 2x - 15 = (x-5)(x+3).$$

For $x \neq -3$,

$$\frac{x+3}{(x-5)(x+3)} = \frac{1}{x-5}.$$

Therefore

$$\boxed{-\frac{1}{8}}.$$

(5) (2 points)

$$\lim_{x \rightarrow -3} \frac{\frac{1}{1-x} - \frac{1}{4}}{x+3}.$$

Combine the fractions in the numerator:

$$\frac{1}{1-x} - \frac{1}{4} = \frac{4 - (1-x)}{4(1-x)} = \frac{x+3}{4(1-x)}.$$

Hence, for $x \neq -3$,

$$\frac{\frac{x+3}{4(1-x)}}{x+3} = \frac{1}{4(1-x)}.$$

Now substitute $x = -3$:

$$\boxed{\frac{1}{16}}.$$